



Maintenance Schedule

Diesel engine

12V4000P63

16V4000P63

20V4000P63

Application group 3A 3B 3C

MS50246/03E

Engine model	kW/cyl.	Application Group
12V4000P63	112,5 kW/cyl.	3A, Continuous operation, unrestricted, ICXN
12V4000P63	130 kW/cyl.	3B, Continuous operation, variable load, ICXN
12V4000P63	130 kW/cyl.	3C, Time-limited operation, ICXN
16V4000P63	112,5 kW/cyl.	3A, Continuous operation, unrestricted, ICXN
16V4000P63	130 kW/cyl.	3B, Continuous operation, variable load, ICXN
16V4000P63	130 kW/cyl.	3C, Time-limited operation, ICXN
20V4000P63	112,3 kW/cyl.	3A, Continuous operation, unrestricted, ICXN
20V4000P63	112,5 kW/cyl.	3A, Continuous operation, unrestricted, ICXN
20V4000P63	130 kW/cyl.	3B, Continuous operation, variable load, ICXN
20V4000P63	130 kW/cyl.	3C, Time-limited operation, ICXN

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product manufactured by Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of operational availability. The maintenance intervals as well as the required maintenance task scopes are based on operational experience and are therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of demanding operational and ambient conditions.

Observance of the specified maintenance intervals is essential to maintain product safety.

The intervals at which the maintenance tasks are to be performed are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule)

QL3: Maintenance work which requires partial disassembly of the product

QL4: Maintenance work which requires complete disassembly of the product

Additional notes on maintenance and preservation:

Maintenance activities stipulated in any documentation provided by third-parties (suppliers and component manufacturers), particularly those relating to individual components and subsystems, may differ from the information specified in this Maintenance Schedule. The Maintenance Schedule provided by Rolls-Royce Solutions shall take precedence in such case.

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

Refer to the applicable Fluids and Lubricants Specifications for details of fluid and lubricant change intervals.

If the product is to remain out of service for a longer period of time, Rolls-Royce Solutions recommends carrying out a monthly test run until the engine has reached its normal operating temperature. If the engine is scheduled to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications.

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Allocation matrix application group 3A, 3B, 3C

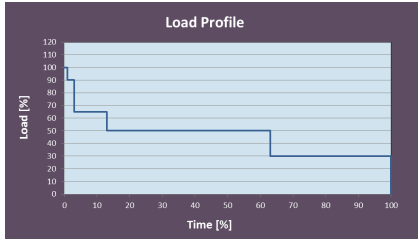
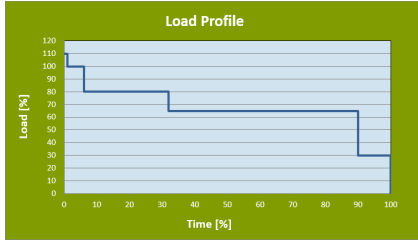
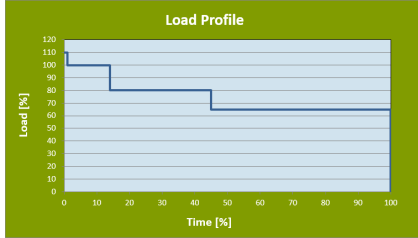
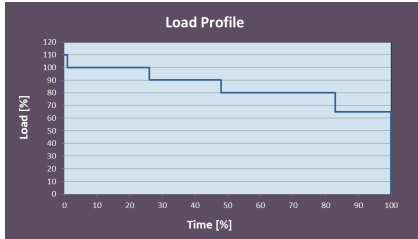
Load profile		Load factor %	Load indicator %	0 to 113 kW/cyl.	114 to 130 kW/cyl.	-	-	
Load %	Time %							
110 ₍₁₀₀₋₁₁₀₎	—	0 to 45	≤ 3	42,000	30,000	—	—	
100 ₍₉₀₋₁₀₀₎	1							
90 ₍₈₀₋₉₀₎	2							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	10							
50 ₍₃₀₋₅₀₎	50							
30 ₍₅₋₃₀₎	37							
Idle speed ₍₀₋₅₎	—	46 to 68	≤ 12	36,000	27,000	—	—	
110 ₍₁₀₀₋₁₁₀₎	1							
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	26							
65 ₍₅₀₋₆₅₎	58							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	10							
Idle speed ₍₀₋₅₎	—	69 to 75	≤ 20	30,000	24,000	—	—	
110 ₍₁₀₀₋₁₁₀₎	1							
100 ₍₉₀₋₁₀₀₎	13							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	31							
65 ₍₅₀₋₆₅₎	55							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle speed ₍₀₋₅₎	—	76 to 85	≤ 40	24,000	21,000	—	—	
110 ₍₁₀₀₋₁₁₀₎	1							
100 ₍₉₀₋₁₀₀₎	25							
90 ₍₈₀₋₉₀₎	22							
80 ₍₆₅₋₈₀₎	35							
65 ₍₅₀₋₆₅₎	17							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle speed ₍₀₋₅₎	—							

Table 2: Allocation matrix application group 3A, 3B, 3C

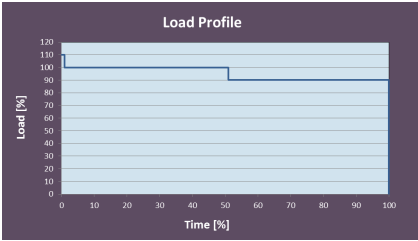
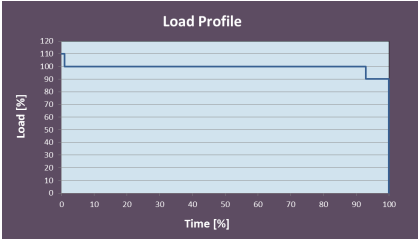
Load profile		Load factor %	Load indicator %	0 to 113 kW/cyl.	114 to 130 kW/cyl.			
Load %	Time %							
110 ₍₁₀₀₋₁₁₀₎	1	86 to 95	≤ 70	21,000	18,000	—	—	
100 ₍₉₀₋₁₀₀₎	50							
90 ₍₈₀₋₉₀₎	49							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle speed ₍₀₋₅₎	—							
110 ₍₁₀₀₋₁₁₀₎	1	96 to 99	≤ 98	18,000	12,000	—	—	
100 ₍₉₀₋₁₀₀₎	92							
90 ₍₈₀₋₉₀₎	7							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle speed ₍₀₋₅₎	—							

Table 3: Allocation matrix application group 3A, 3B, 3C

1.4 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0001	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.5 Maintenance Schedule Matrix 42000 HRS

500 to 15000 Operating hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00290	1 DAY																														
WM00291	1 DAY																														
WM00174	1 WK																														
WM00201	1 MON																														
WM00296	1 MON																														
WM00321	1 MON																														
WM00639	1 MON																														
WM00881	1 MON																														
WM00126	1 YR																														
WM00128	1 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00066	9 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00245	3 MON		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00084	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00085	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00050	1 YR						X						X						X						X						X
WM00136	1 YR						X						X						X						X						X
WM00076	3 YR						X						X						X						X						X
WM00124	3 YR						X						X						X						X						X
WM00052	18 YR						X						X						X						X						X
WM00094	18 YR		X				X						X						X						X						X
WM00119	3 YR												X												X						
WM00089	18 YR																					X									
WM00090	18 YR																					X									
WM00098	18 YR																					X									
WM00110	18 YR																					X									
WM00775	18 YR																												X		
WM00091	18 YR																														
WM00087	18 YR																														
WM00976	18 YR																												X		
WM00008	6 YR																														
WM00065	9 YR																														
WM00054	18 YR																														
DAY = Days																															
WK = Weeks																															
MON = Months																															
YR = Years																															
HRS = Hours																															

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00056	18 YR																														
WM00062	18 YR																														
WM00093	18 YR																														
WM00096	18 YR																														
WM00164	18 YR																														
WM00921	18 YR																														
WM00970	18 YR																														
WM00971	18 YR																														
WM00973	18 YR																														
WM00974	18 YR																														
WM00986	18 YR																														
WM00005	6 YR																														
WM00006	6 YR																														
WM00011	18 YR																														
WM00045	18 YR																														
WM00061	18 YR																														
WM00072	18 YR																														
WM00079	18 YR																														
WM00088	18 YR																														
WM00092	18 YR																														
WM00099	18 YR																														
WM00100	18 YR																														
WM00104	18 YR																														
WM00108	18 YR																														
WM00127	18 YR																														
WM00137	18 YR																														
WM00353	18 YR																														
WM00439	18 YR																														
WM00925	18 YR																														
WM00980	18 YR																														
WM00001	18 YR																														
WM00002	18 YR																														
WM00070	18 YR																														
WM00095	18 YR																														
WM00103	18 YR																														
WM00111	18 YR																														
WM01941	1 MON																														
WM00978	18 YR																														
WM00222	1 YR																														
WM00148	3 YR																														
WM00979	18 YR																														
WM00228	6 MON	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00190	1 YR																														
WM00191	1 YR																														
DAY = Days																															
WK = Weeks																															
MON = Months																															
YR = Years																															
HRS = Hours																															

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